Faris ElBaz

U.S Citizen | Software Engineer | AI & High-Performance Systems

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Education

The Knowledge Hub Universities (Coventry University Branch) May 2026
Bachelor of Science in Computer Science *Cairo, Egypt*
- **GPA: First class honors** | IELTS 8.5/9.0 | **Course Representative** (2025–2026): Academic liaison for 30+ students
- **Relevant Coursework:** Parallel Programming, Deep Learning, State-Space Models (SSMs), Distributed Systems, Operating Systems

Technical Skills

Languages: Python, C++, CUDA, C#, SQL, JavaScript, Dart
AI & Machine Learning: PyTorch, LightGBM, scikit-learn, Transformers, Mamba (SSMs), Captum, XAI
Data & Systems: DuckDB, Polars, OpenMP, CUDA Kernel Optimization, Memory Coalescing, POSIX Concurrency
Backend & Tooling: FastAPI, Docker, Linux, Git, Node.js, Express, REST APIs, Firebase, MongoDB

Professional Experience

FlyRank August 2026 – October 2026
Machine Learning Engineering Intern *Remote*
- Architected an out-of-domain content trajectory prediction pipeline in **DuckDB**, **Polars**, and **LightGBM** across an 81.8M-row star-schema telemetry warehouse.
- Engineered 14-day rolling demand velocity, SERP rank volatility, and CTR deficit features under strict cutoff boundaries to prevent temporal lookahead bias and target leakage.
- Implemented a 5-fold **GroupKFold** validation scheme grouped by client tenant hash to eliminate cross-domain leakage, boosting PR-AUC from **0.2728** (heuristic baseline) to **0.5613** (**+105.8% relative lift**) with a **0.9175** ROC-AUC.
- Audited model failure modes (zero-click SERPs, exogenous demand shocks) and deployed a deterministic 5-tier action playbook and public research paper prioritizing high-upside editorial interventions.

Omdena October 2025 – December 2025
AI Backend Engineering Intern *Remote*
- Engineered a high-throughput context summarizer service in **FastAPI** to optimize LLM agent memory and eliminate redundant context payload transfers.
- Built asynchronous API endpoints and caching logic that cut agent token consumption by **35%** and reduced context retrieval latency by **180ms**.
- Hardened distributed agent pipelines across edge environments by implementing exponential backoff retries and payload validation schemas.

Technical Projects

Deep Mendelian Inference Engine | ***PyTorch, Mamba, Captum*** [GitHub] Graduation Project
- Achieved **97% classification accuracy** and **0.85 F1 score** by architecting a hybrid Transformer and Mamba (SSM) pipeline with $O(N)$ linear sequence scaling.
- Expanded training data by **1,000%** via a synthetic Mendelian-consistent trio generator from the 1000 Genomes Project dataset.
- Integrated Captum for SHAP-based feature attribution to provide interpretable, clinical-grade variant pathogenic risk assessments.

Optimization of K-means Clustering | ***C++, CUDA, OpenMP*** [GitHub] 2026
- Achieved a **24.6x speedup** (3.2s → 0.13s) over sequential CPU baseline via custom CUDA thread-block centroid decomposition.
- Eliminated global GPU memory latency bottlenecks by implementing 2D shared memory tiling and coalesced memory access patterns.
- Benchmarked OpenMP multi-core CPU scaling against CUDA kernel dispatch overhead across high-dimensional feature spaces.

Drill To Glory: 2D Strategy Deckbuilder | ***Unity, C#, Firebase*** [GitHub] 2026
- Programmed turn-based game mechanics using State and Singleton design patterns for deterministic multi-state game-loop execution.
- Architected real-time backend state sync with Firebase Firestore, Auth, and Cloud Functions with transactional validation rules.

Role-Based Secure Healthcare Application | ***Node.js, Express, React, MongoDB*** [GitHub] 2025
- Architected a medical workflow backend implementing double-submit CSRF protection, IP rate limiting, and regex NoSQL injection sanitization.
- Engineered session management utilizing TOTP MFA, JWT role guards, and an in-memory token-revocation blacklist.

Certifications & Honors

Top 5 Placement – Annual Computing Exposition, The Knowledge Hub / Coventry University

AI Innovation Challenge Award – Collaborative AI Engineering Architecture, Omdena

Specializations – Supervised Machine Learning & Deep Learning, DeepLearning.AI / Stanford Online